ADRIAN LEONG TAT WEI

PASSIONATE DATA SCIENTIST

CONTACT

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Klang Valley, Selangor.

PROGRAMMING SKILLSET

- Python
- R
- C/C++
- SQL
- Pytorch
- Tensorflow
- Caffe
- OpenCV
- Spark
- Hadoop
- NumPy
- Pandas
- Computer Vision

PROFILE

Engineer focused on making life easier through innovative tech and AI. Passionate about solving real-world problems. Eager to contribute to the next big breakthrough that transforms how people live, work, and interact with the world.

WORK EXPERIENCE

Myrmidon Labs - Singapore APR 2022 - SEP 2023

Robotics & AI Engineer

- **Acoustic Data Transmission:**
  - Developed sound-based communication over solid barriers (e.g., metal doors) where Wi-Fi fails.
  - Enabled RF-free connectivity for secure/industrial environments, reducing wiring costs and improving reliability in metal-heavy infrastructure.
  - ***Tools: ESP32, C++, I2S***

- **Thermal Imaging People Detection:**
  - Built a thermal sensing system (8×8 to 64×64 px) to detect and localize people using low-resolution data.
  - Developed a multi-purpose thermal camera system that ensures privacy-conscious occupancy monitoring (e.g., in hospitals or restrooms) while also doubling as an effective tool for detecting equipment overheating and fire hazards.
  - ***Tools: ESP32, C++, I2C***

- **Mandai Zoo Interactive System:**
  - Designed a scheduling system to trigger audio and LED patterns via a web interface.
  - Automated animal cue training and reduced manual intervention, reducing labour needed for smooth exhibit operation and improving customer experience.
  - ***Tools: ESP32, C++, UART, SPI, embedded HTTP server***

Azure Innovations - KL SEP 2020 - FEB 2022

AI & Machine Learning Engineer

- **Enhanced People Detection System:**
  - Research and develop computer vision, ML, and AI algorithms tailored to client needs
  - Design and optimize models for production
  - Adapt models to handle ambiguous or imperfect data
  - Improve speed and efficiency with minimal accuracy loss
  - ***Tools: Python, OpenCV, TensorFlow, PyTorch, Caffe***

EDUCATION

Master of Data Science Graduating July 2026

Monash University Sunway

Bachelor of Electrical Engineering (HONS) 2016 to 2019

Monash University Clayton, Australia